

(2)

The transmission system and the distribution centres, it supplies the electrical power from the sources (generation power plants) to the distribution system (load centres). Figure 1 shows the structure of a typical power system structure.

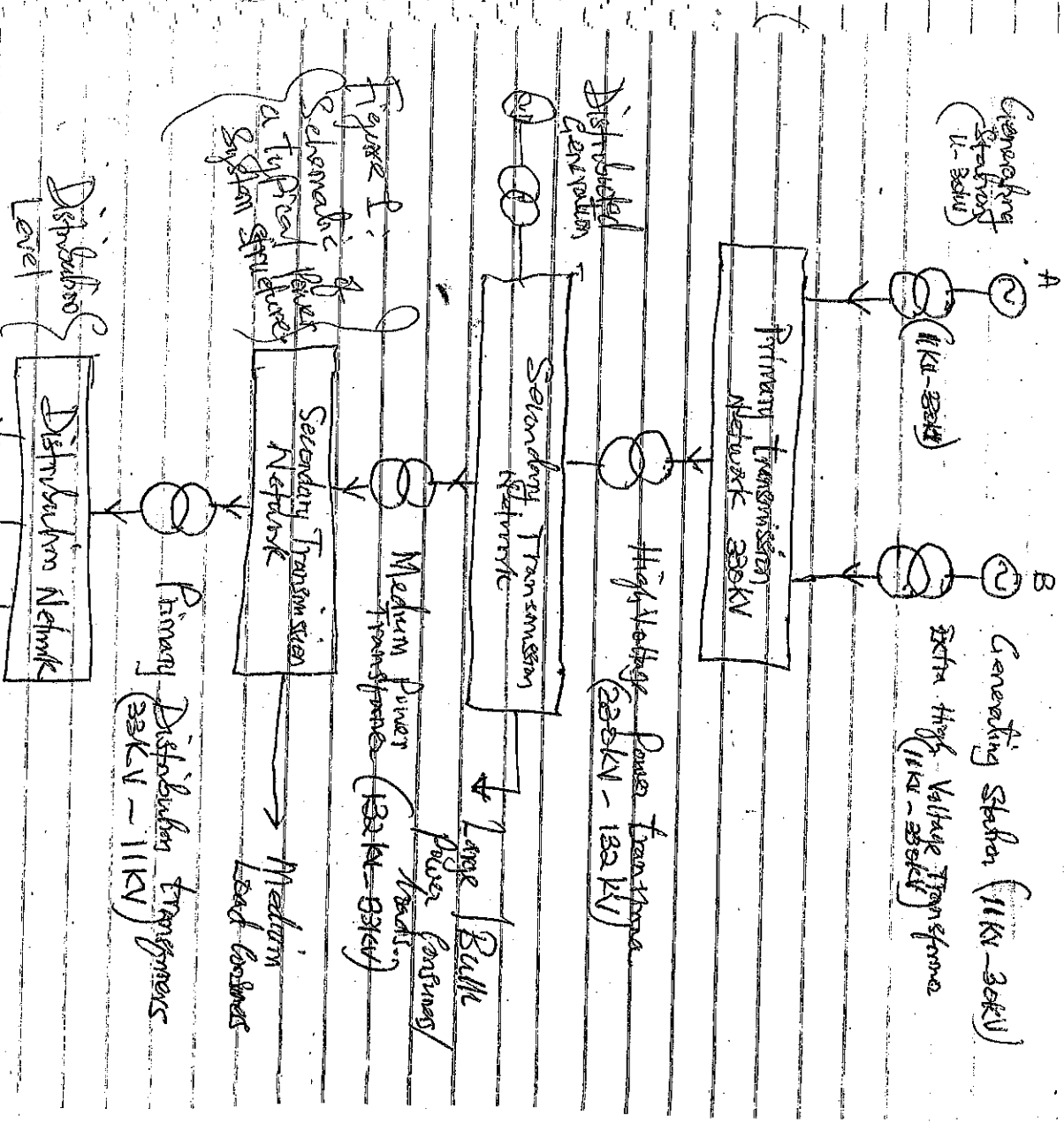


Figure 1: Schematic view of a typical power system structure

In Nigeria electricity is generated at voltage between (22kV - 33kV) at the generation station which is then stepped-up through an extra high-voltage power

ELECTRICAL

POWER SYSTEM II EE-232

A) SOURCES OF ELECTRICAL ENERGY

Electrical energy is a product of primary energy sources mainly fossil fuels and solar energy derivatives. Generally the various sources of electrical energy can be broadly classified into two major types namely: the conventional energy sources and the non-conventional energy sources.

X Conventional energy sources are also referred to as the non-renewable energy sources since the cannot easily be replaced when replenished. This form of energy sources cannot be easily regenerated to meet up with demand. They are made from dead decayed organic matter. Examples includes: coal, oil, natural gas and the nuclear energy sources. The non-conventional energy sources are the renewable energy sources, they are in-exhaustible energy source they are continuously being replenished through natural processes. These form of energy sources are often referred to as clean energy because they do not pollute the environment. These form of energy are either from solar direct sources or solar indirect i.e they are solar derivatives. Examples of non-conventional energy sources are: Wind, Solar, Biomass, Biogas, Ocean tidal/thermal, geothermal, Fuel cells etc.

B) POWER SYSTEM STRUCTURE

The electrical power system is made of an interconnection of three major components or links: the generation system, the transmission system and the distribution system. The interconnection of these three elements is known as a grid or electrical grid. The network which interconnects the generation units